

A Cross Country Analysis on the Impact of Export Growth on GDP Growth

1. Abstract

This paper investigates the impact of export growth on GDP growth across a population of 34 countries over 20 years, utilizing comprehensive macroeconomic data sourced from the International Monetary Fund (IMF). The analysis employs panel data regression and correlation techniques to evaluate the strength of the export-GDP relationship while identifying and controlling for country-specific effects and global economic trends. The findings reveal a positive correlation between export performance and GDP growth. The data analysis and trends support the hypothesis that export growth contributes to overall economic development. While exports are not the only determinant of growth, their influence remains crucial for emerging and export-oriented economies.

2. Introduction

2.1 Overview

The relationship between exports and economic growth has been extensively studied in the economic literature. Trade is one of the key drivers of economic growth. However, the strength and nature of this relationship vary on the basis of a country's domestic and international development stage (Yaya Keho, 2018). The export-driven growth hypothesis reflects the relationship between exports and economic growth, emphasising the fact that exports help in promoting economic growth through improved productivity, diffusion of technological advancements, efficient allocation of resources, and economies of scale. These positive externalities promote economic growth (Mishra, n.d.).

Based on the above-mentioned factors, many developing countries have adopted export-oriented policies as part of their development strategies. Additionally, outward trade orientation has been identified as a key factor behind the remarkable success of the East Asian tiger economies over the past three decades (World Bank, 1993; Stiglitz, 1996). As a result, exports are widely regarded as an important driver of economic growth.

However, there exists ambiguity regarding whether exports consistently lead to economic growth across all countries. It may be the opposite, ie. exports are driven by economic growth or the two variables both affect one another (Dutt and Ghosh, 1994).

It is due to these contrasting findings that countries still debate whether to open up their economies, despite its potential challenges to the domestic economies, or primarily focus on driving economic growth. By employing cross-country analysis, this study contributes to the broader literature on trade and growth, offering empirical insight into whether export performance correlates with national economic output.

This paper explores the relationship between export and GDP growth across a set of 34 countries, including developed, developing, and emerging nations, over the 20-year period from 2004 to 2024. By analyzing this varied dataset along with a sample derived from it, the paper aims to highlight both global development patterns in how export performance correlates with economic development. We aim to understand whether increases in exports translate into higher economic growth uniformly across different stages of economic development. The remainder of this paper is organized into five sections. Section 2 discusses the methodology and empirical model. Section 3 discusses the data and related descriptive statistics. Section 4 presents the results of our study, and Section 5 presents the concluding remarks.

2.2 Theoretical Perspectives

The Keynesian theory emphasizes the role of aggregate demand in driving economic output. This suggests that increased exports boost national income by raising external demand for domestically produced goods and services.

The Neoclassical perspective focuses on long-run growth driven by factors including capital accumulation, efficiency of resources etc. This suggests that exports contribute to growth by enabling economies to specialize on the basis of comparative advantage.

3. Methodology

3.1 Data Sources and Preparation

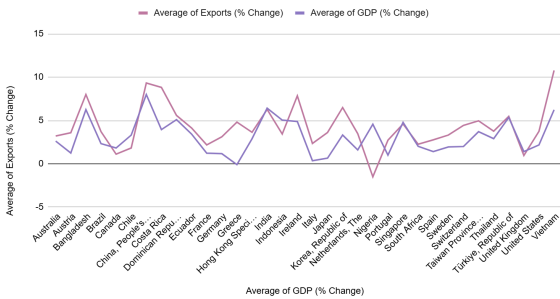
The raw data sheet contains statistics on two macroeconomic variables which are - Volume of exports of goods (% change) and GDP at constant prices (% change). The author has chosen 34 countries over a period of 20 years, representing a wide range of income levels, geographic regions, and stages of economic development. The data was sourced from the World Economic Outlook database published by the International Monetary Fund (IMF), which is a credible data source. GDP was chosen as the dependent variable, and exports is the independent variable. The dataset was randomly selected from a list of 126 countries hence, issues of sample selection bias are unlikely to arise. Following the collection, the data was then cleaned manually. The clean datasheet represents the cleaned version of the raw dataset. The cleaning process involved rearranging the dataset in designated rows and columns and retaining significant outliers linked to major economic events for analytical purposes. To analyse the distribution of the population data and identify outliers, the author has made use of box plots, histograms, and scatterplots. For the purpose of hypothesis testing and deeper analysis, a random sample consisting of 10 countries over a 10-year period was selected from the population dataset. This smaller sample facilitates detailed statistical examination while maintaining representativeness.

4. Descriptive Statistics

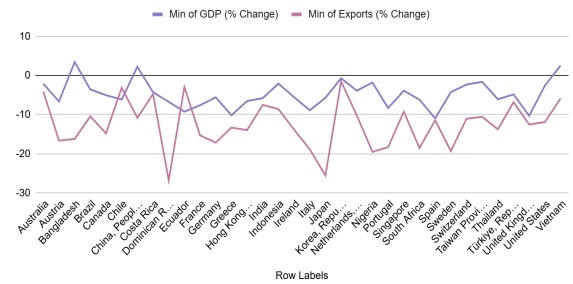
4.1 Basic Descriptive Statistics

LABELS	MEAN	MEDIAN	MAX	MIN	RANGE	STD DEV	Q1	Q3	IQR
GDP	3.09	2.85	24.62	-10.94	35.56	3.68	1.35	5.38	4.03
EXPORTS	4.26	3.73	95.95	-26.79	122.74	8.67	0.51	8.08	2.70

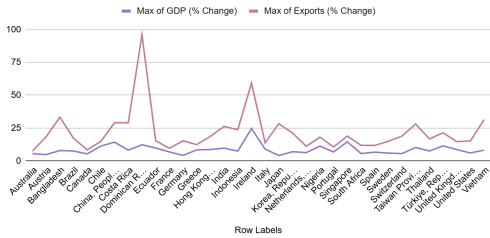
Average of Exports (% Change) vs. Average of GDP (% Change)



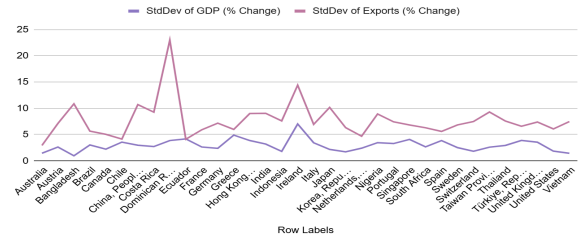
Min of GDP (% Change) and Min of Exports (% Change)



Max of GDP (% Change) and Max of Exports (% Change)



StdDev of GDP (% Change) and StdDev of Exports (% Change)



4.2 Descriptive Statistical Analysis of Export Growth and GDP Growth

Measures of Central Tendency

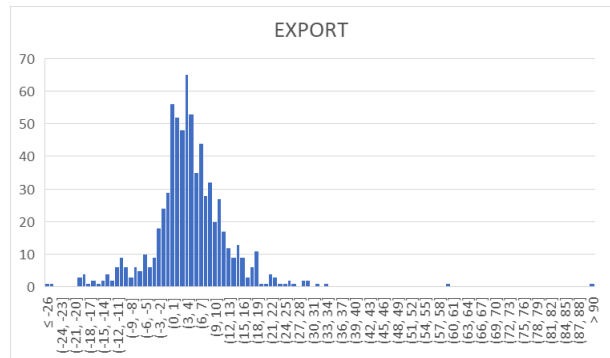
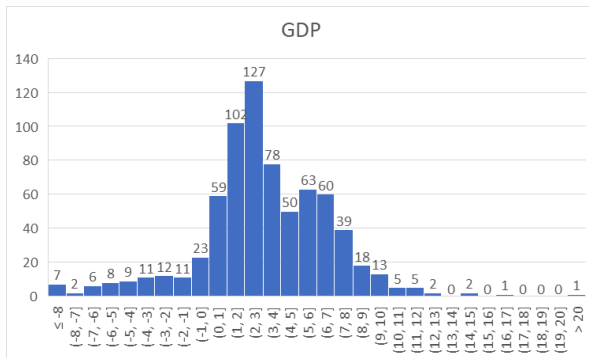


Fig 1 & 2

The central tendency of the two variables indicates observable differences in the distribution. The mean and median GDP growth rates are closely aligned at approximately 3.09% and 2.85%, respectively. This suggests a relatively symmetrical distribution, indicating that most countries experienced steady and moderate economic growth over the observation period.

In contrast, export growth shows a higher mean of 4.26% compared to a median of 3.73%. The slight gap between the mean and median implies a slightly right-skewed distribution, where a few countries with exceptionally high export growth rates are pulling the average upward. This suggests that while most countries experienced moderate export growth, a small number significantly outperformed the rest, influencing the overall average.

5. Variability

5.1 Overall Variability in GDP Growth and Export Performance

Export growth shows higher variability across countries, with an overall variance of 75.129, compared to a variance of 13.56 for GDP growth. This difference indicates that changes in exports face greater fluctuations as compared to changes in GDP. This may be due to the differences in the country's levels of trade openness, global market exposure, or dependence on volatile commodities. GDP growth, while more stable overall, still spans a wide range, which can be linked to global financial crises, economic reforms, or major macroeconomic transitions.

5.2 GDP Growth

Countries exhibiting high variability in GDP growth, which is defined here as a variance exceeding 20 %, include Ireland (49.33 %) and Greece (24.08 %). Ireland's variance largely reflects its fiscal and financial restructuring in the wake of the 2008–09 financial crisis, followed by a rapid, export-led recovery. Greece, on the other hand, endured severe setbacks during both the global financial crisis and more recent pandemic-related disruptions.

By contrast, a number of economies show very low GDP growth variance (below 5 %), notably Bangladesh (0.94 %), Australia (2.13 %), Vietnam (2.08 %), the United States (3.45 %), and the Republic of Korea (2.94 %). These relatively stable growth profiles can be attributed to robust policy frameworks and institutional strength, as in the case of the U.S. and Australia, and to gradual, incremental development strategies, which in emerging markets like Bangladesh and Vietnam have helped to buffer against abrupt shocks.

5.3 Export Growth

Some countries again stand out for exceptionally high variance (above 95 %) include Bangladesh (302.56 %), Indonesia (266.78 %), China (207.90 %), Italy (134.63 %), and Thailand (96.10 %). Such volatility is characteristic of export-driven economies that depend heavily on a narrow range of commodities brackets, making them particularly sensitive to commodity-price and fluctuations. Conversely, nations with relatively low export variance (below 35 %), including South Africa (8.69 %), Brazil (16.98 %), Korea (22.08 %), Turkey (31.47 %), and Spain (32.06 %), benefit from diversified export baskets and stable

long-term trade relationships, which together help to dampen the impact of sector-specific or geopolitical disturbances.

Row Labels	Variance of GDP (% Change)	Variance of Exports (% Change)
Australia	2.133234162	8.691931629
Austria	6.950063714	50.80554195
Bangladesh	0.94344699	117.9332935
Brazil	9.184861957	32.05550793
Canada	5.000440114	25.52556755
Chile	12.73140365	17.22790376
People's Republic of China	8.856357262	114.8946658
Costa Rica	7.4296483	85.95246131
Dominican Republic	15.00180239	528.8100877
Ecuador	17.53381506	16.98065125
France	6.995634148	35.03576819
Germany	5.727596948	51.62985633
Greece	24.08910083	35.92436196
Hong Kong	15.04693956	81.18963126
India	10.26105023	81.90883856
Indonesia	3.247386157	57.86725021
Ireland	49.33178091	207.9027278
Italy	11.86813066	48.34085196
Japan	4.764394914	103.5711201
Republic of Korea	2.944909629	40.08428056
The Netherlands	5.84144869	22.08224336
Nigeria	12.00579841	79.69674
Portugal	10.87433823	55.33357393
Singapore	16.67629469	46.76492916
South Africa	7.145076062	39.73687046
Spain	14.90936295	31.46788156
Sweden	6.387791757	46.80301001
Switzerland	3.345752062	56.02097106
Taiwan Province of China	6.813340048	86.58001199
Thailand	8.684883557	57.52556975
Republic of Türkiye	15.19224803	43.36196113
United Kingdom	12.62157681	54.69824365
United States	3.447448962	37.04941669
Vietnam	2.079074062	56.08315553
Grand Total	13.56475946	75.12947078

6. Skewness

6.1 GDP Growth

The mean and median GDP growth rates are close (Mean = 3.09%, Median = 2.85%), highlighting that the distribution is nearly symmetrical. This is supported by the skewness value of -0.13, suggesting a leftward skew of the graph. Since the range is 35.56 values, and the standard deviation is 3.68, there exists substantial volatility. The interquartile range (IQR) is 4.03, which indicates moderate dispersion around the middle 50% of data. These suggest that while most GDP growth values are stable and symmetrical, occasional economic downturns or recessions contribute to temporary dips, resulting in a slightly extended left tail. Nonetheless, the overall distribution remains balanced.

6.2 Export Growth

The export growth rate distribution is strongly right-skewed, as evidenced by a skewness value of 1.9. This is because certain extreme export growth values significantly pull the distribution to the right. The mean (4.26%) is significantly higher than the median (3.37%), reinforcing the right-skewed nature of the distribution. The range is 122.74 values, and the standard deviation is 8.67, indicating high variability in export performance. The IQR is 2.70, suggesting that most countries are within the central 50% of data. However, a few countries experienced unusually high spikes in export growth. These outliers skew the distribution and raise the average.

7. Outliers

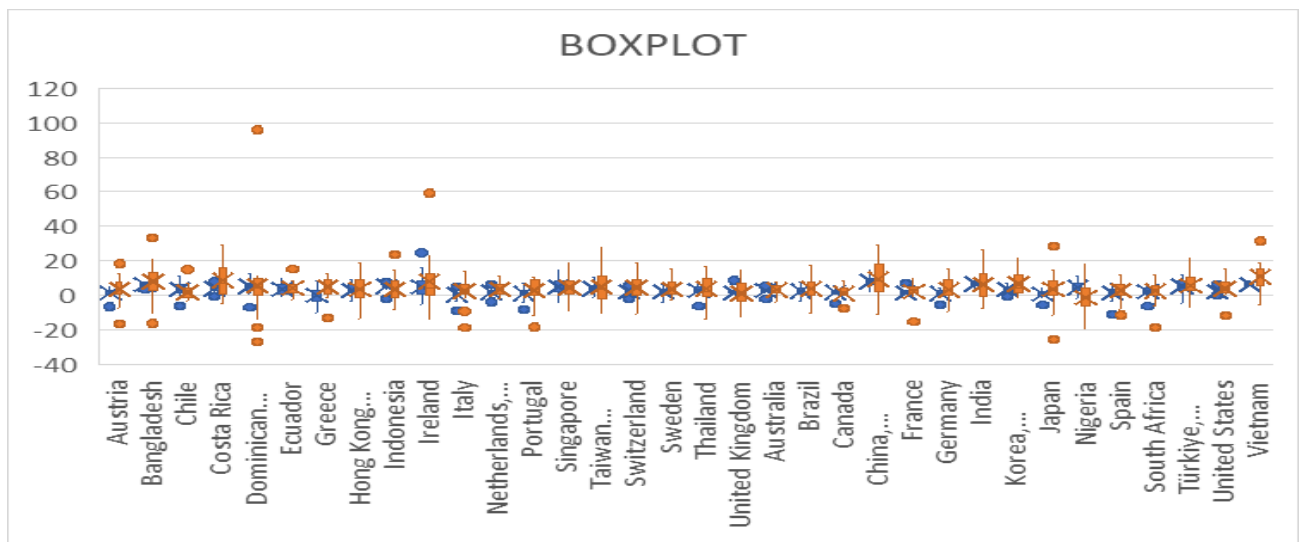


Fig 3

The author has made use of statistical analysis using scatterplots and histograms to highlight certain data points that stood out for exhibiting extreme values in either Exports Growth or GDP Growth.

Metric	Country	Value (%)	Year	Reason
Exports	Dominican Republic	95.95	2010	Economic growth, trade expansion
Exports	Dominican Republic	-26.79	2009	2008 Global crisis
Exports	Ireland	59.15	2015	Multinational Exports
Exports	Bangladesh	33.25	2011	Manufacturing Industry
Exports	Vietnam	31.45	2004	Manufacturing Industry
Exports	Japan	-25.54	2009	2008 Global crisis
Exports	Nigeria	-19.50	2021	Pandemic
Exports	Sweden	-19.28	2009	2008 Global crisis
GDP	Ireland	24.62	2015	Economic growth and reforms
GDP	Austria	-6.63	2020	Pandemic
GDP	Spain	-10.94	2020	Pandemic

Outliers were excluded from our analysis to prevent these extreme values from unduly skewing summary statistics and distorting inferential results since outliers can inflate measures of dispersion, as seen in the graphs, their presence may mask the underlying patterns that most countries exhibit. By removing these observations, we ensure that estimates of central tendency and variability more accurately reflect the typical range of country performance. Restoring a narrower confidence interval to reveal the underlying relationship between GDP growth and export performance more clearly, is an essential condition for robust hypothesis testing. Although these data points represent historically significant episodes, their magnitude is not representative of general economic behavior and would distort any general inference.

8. Regression Analysis

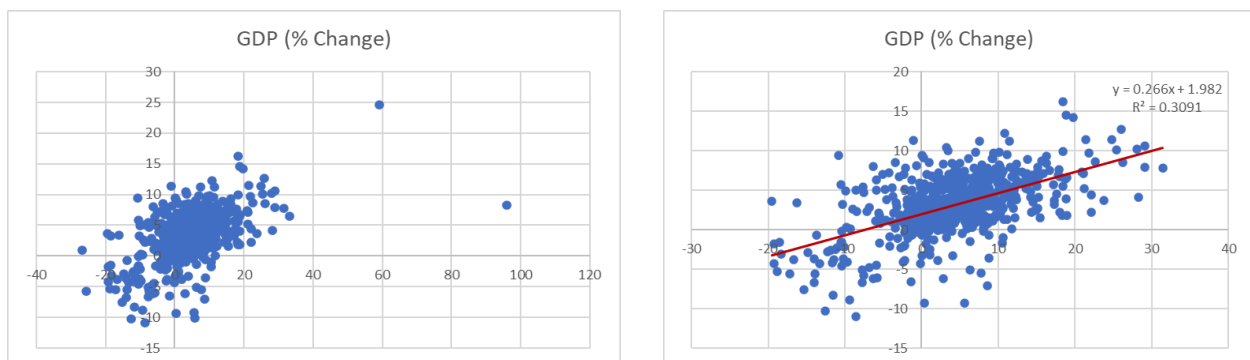


Fig 4 & 5

The scatter plot reveals a clear, positive relationship between export growth (independent variable) and GDP growth (dependent variable) across 34 countries. The fitted regression line, with equation $y = 0.266x$

+ 1.982 and an R^2 value of 0.3091, indicates that approximately 31% of the variability in GDP growth can be attributed to differences in export performance. In Figure A (with outliers), the outliers pulled the regression line toward the tails. Once those outliers are removed (Figure B), the fitted line remains positively sloped, confirming that higher export growth generally predicts stronger GDP growth, but it now cuts more cleanly through the main cluster of country observations. The data points exhibit tighter dispersion around the line and the explained variance rises, producing a more precise, reliable depiction of the typical export-to-GDP relationship and improving both visual clarity and statistical interpretability.

Source SS	df MS	Number of obs =	707	
		F(1, 705) =	315.37	
Model 2754.56059	1 2754.56059	Prob > F =	0	
Residual 6157.70911	705 8.73433916	R-squared =	0.3091	
		Adj R-squared =	0.3081	
Total 8912.2697	706 12.6236115	Root MSE =	2.9554	
GDPChange	Coef.	Std. Err. t	P>t [95% Conf. Interval]	
ExportsChange	0.266047	.0149812 17.76	0.000 .2366338	0.2954601
_cons	1.982027	.1271966 15.58	0.000 1.732298	2.231757

The author ran a simple regression using 707 observations to see how export growth affects GDP growth. The result shows that for every 1 %age-point increase in export growth, GDP growth rises by about 0.27 %age points on average. The table highlights that, if export growth were zero, GDP would still be expected to grow by roughly 1.98 %. With an R^2 of 0.309 (adjusted $R^2 = 0.308$), export performance explains about 31 % of the variation in GDP growth and the remaining 69 % stems from other factors not included in this simple model. Overall, the table confirms that export growth is a highly significant predictor of GDP growth.

9. Hypothesis Testing

9.1 Research Hypothesis

This paper aims to study the extent to which international trade performance, as a measure of export growth, influences domestic economic growth (GDP) across a diverse sample of countries. To prove so, the following hypothesis are proposed by the author:

Null Hypothesis (H_0): There is no statistically significant relationship between export growth and GDP growth across countries.

Alternative Hypothesis (H_1): Export growth has a positive and statistically significant effect on GDP growth across countries.

This hypothesis is grounded in the theories of export-led growth models, which claim that increased participation in international trade can stimulate domestic production, enhance productivity, and ultimately accelerate economic growth.

9.2 Sampling Methodology

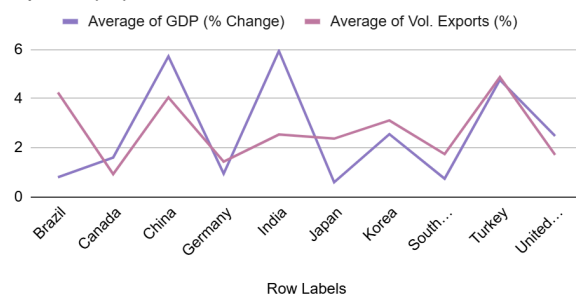
A simple random sampling technique was utilised. Using the rand() function in Excel which randomises a selected range, a selection of 10 countries were drawn from the original dataset. This method enabled an unbiased extraction of a sample set. The selected countries include Korea, South Africa, Japan, Brazil, Turkey, India, Canada, China, Germany, and the United States. They represent a broad spectrum of geographical regions and income classifications. This diversity helps preserve heterogeneity in economic performance within the sample. Data spanning a 10-year period for these countries was utilized to conduct the regression analysis in the subsequent sections.

9.3 Descriptive Statistics (Summary)

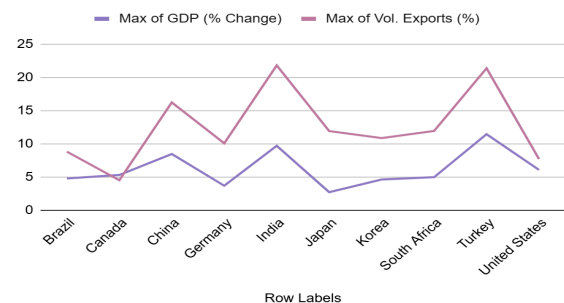
For the sample, the mean, standard deviation, max, min, and quartiles are computed.

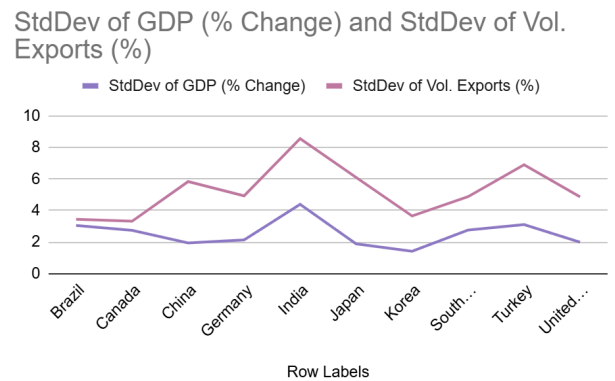
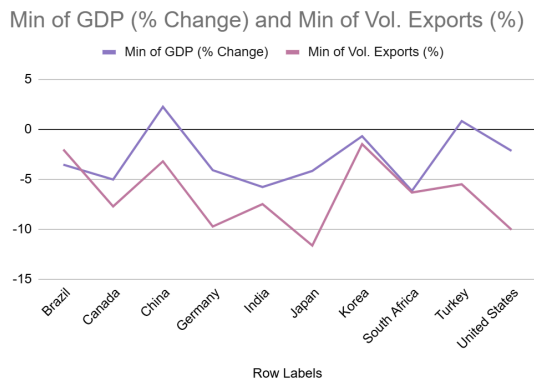
LABELS	MEAN	MEDIAN	MAX	MIN	RANGE	STD DEV	Q1	Q3	IQR
GDP	2.62	2.55	11.44	-6.17	17.61	3.24	1.15	4.65	3.50
EXPORTS	2.70	2.84	21.79	-11.64	33.43	5.37	0.30	4.79	4.49

Average of GDP (% Change) and Average of Vol. Exports (%)



Max of GDP (% Change) and Max of Vol. Exports (%)





9.4 Descriptive Statistics

9.4.1 Measures of Central Tendency

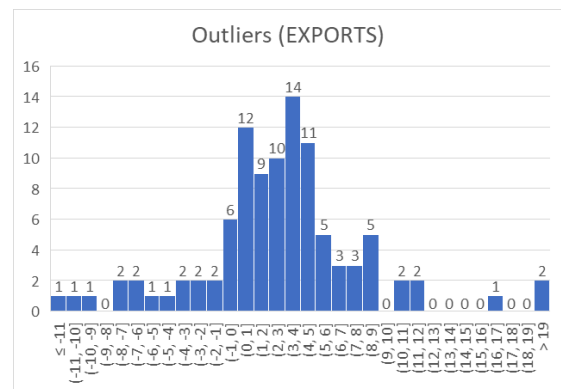
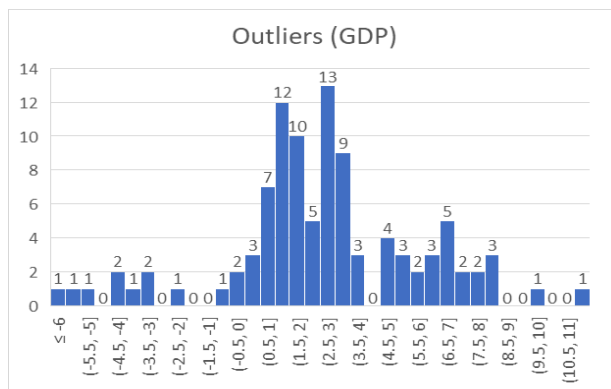


Fig 6 & 7

The GDP growth variable shows a mean of 2.62% and a median of 2.55%, which indicates a fairly symmetrical distribution with moderate growth across most countries. The range spans from -6.17% to 11.44%, and a standard deviation of 3.24% suggesting a moderate dispersion around the mean. The interquartile range (IQR) of 3.50, highlights a relatively consistent pattern of economic expansion.

On the other hand, export growth displays a slightly higher mean of 2.70% and a median of 2.84%, implying a mildly left-skewed distribution where more countries had export growth above the average. However, export data is far more dispersed, as indicated by a standard deviation of 5.37% and a broader range stretching from -11.64% to 21.79%. The IQR of 4.49 further underscores the variability in export performance across countries. From these patterns we can suggest that while GDP growth is more evenly distributed, export growth tends to fluctuate more widely, thus reflecting a heterogeneous trade dynamics among our given sample of nations.

9.4.2 Distribution and Variability

Row Labels	Var of GDP (% Change)	Var of Vol. Exports (%)
Brazil	9.29	11.82
Canada	7.49	11.01
China	3.76	34.04
Germany	4.54	24.30
India	19.27	73.25
Japan	3.54	37.21
Korea	1.99	13.32
South Africa	7.60	23.80
Turkey	9.64	47.57
United States	3.94	23.62
Grand Total	10.51	28.88

Overall Variability in GDP Growth and Export Performance

The variance analysis of the sampled countries highlights a consistent trend observed in broader datasets ie. export growth displays greater volatility than GDP growth. Across the 10 selected countries, the average variance in export performance stands at 28.88%, compared to a lower GDP growth variance of 10.51%.

GDP Growth

Countries such as India (19.27%) exhibit relatively higher GDP volatility, possibly due to economic reforms or exposure to external shocks. In contrast, Korea (1.99%) and Japan (3.54%) display far more stable GDP growth patterns, reflecting stable economic institutions and firm policy implementation.

Export Growth

Even in export growth, India exhibits notably high export variance (73.25%), driven by its reliance on specific commodity baskets for trade such as manufacturing and exposure to global demand cycles. Countries like China and Japan also show relatively high export volatility (34.04% and 37.21% respectively), as a result of their strong participation in global trade. In contrast, the United States and Germany maintain lower variance in both indicators, indicative of diversified economies with strong institutional frameworks.

9.4.3 Skewness

GDP Growth

The distribution of GDP growth in the sample is relatively balanced, with the mean (2.62%) and median (2.55%) being nearly identical. This alignment suggests a symmetrical distribution, which is supported by a skewness of -0.1768. These numbers suggest that GDP growth among our sample countries is nearly even distributed with a minimal leftward skewness.

Export Growth

Export growth in the sample highlights mild right skewness of 0.48, as the mean (2.70%) is slightly less than the median (2.84%), although the difference is minimal.

9.4.4 Outliers

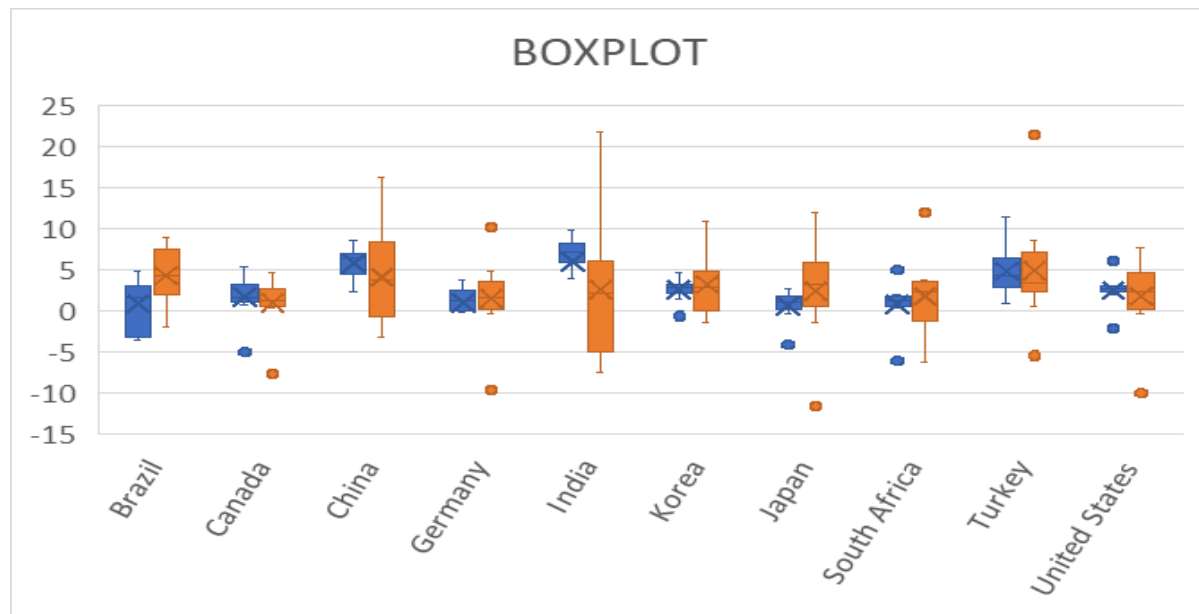


Fig 8

The author has made use of statistical analysis using scatterplots and histograms to highlight certain data points that stood out for exhibiting extreme values in either Exports Growth or GDP Growth.

Metric	Country	Value (%)	Year	Reason
Exports	Turkey	21.368	2021	Post Pandemic Recovery
Exports	India	21.793	2021	Post Pandemic Recovery
Exports	China	16.22	2021	Post Pandemic Recovery
Exports	Japan	-11.637	2020	Pandemic
Exports	Germany	-9.731	2023	Pandemic
Exports	USA	-10.052	2020	Pandemic
Exports	Canada	-7.723	2020	Pandemic
GDP	South Africa	-6.169	2020	Pandemic

Similar to the reasoning for removing outliers in our population data, the outliers were excluded from our sample analysis to prevent these extreme values from skewing summary statistics and distorting inferential results since outliers can skew the measures of dispersion as seen in the graphs, and their

presence in the dataset might mask the underlying patterns that most countries exhibit. By removing these observations, we ensure that estimates of central tendency and variability more accurately reflect the typical range of country performance. Restoring a narrower confidence interval to reveal the underlying relationship between GDP growth and export performance more clearly, is an essential condition for robust hypothesis testing. Although these data points represent historically significant episodes, their magnitude is not representative of general economic behavior and would distort any general inference.

9.4.5 Economic Implications of the Distribution

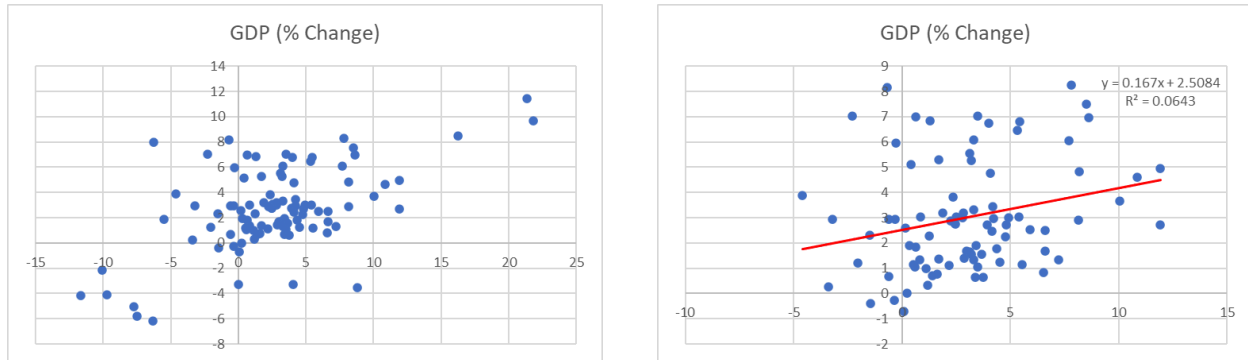


Fig 9 & 10

The scatter plot reveals a clear, positive relationship between export growth (independent variable) and GDP growth (dependent variable) across 34 countries. The fitted regression line, with equation $y = 0.167x + 2.5084$, with a low R^2 value of 0.0643, indicating that only about 6.4% of the variation in GDP growth can be attributed to differences in export performance. In Figure A (with outliers), the outliers pulled the regression line toward the tails.

Source	SS	df MS Number of obs =	100
		F(1, 98) =	44.31
Model	324.08823	1 324.08823 Prob > F =	0
Residual	716.73496	98 7.31362204 R-squared =	0.3114
		Adj R-squared =	0.3044
Total	1040.82319	99 10.5133656 Root MSE =	2.7044
GDPChange	Coef.	Std. Err. t P>t [95% Conf.	Interval]
VolExports	0.3366967	.0505794 6.66 0.000 .2363235	0.4370698
_cons	1.705247	.3030239 5.63 0.000 1.103906	2.306588

The author ran a simple regression using 100 observations to examine how export growth affects GDP growth. The results indicate that for every 1% increase in export growth, GDP growth increases by approximately 0.34%. With an R^2 of 0.311, it suggests that export performance makes up roughly 31% of the variation in GDP growth, while the remaining 69% is influenced by other factors. The coefficient on export growth is highly statistically significant ($p < 0.001$), confirming that export growth is a strong predictor of GDP growth in the sample.

9.4.6 Correlation

(obs=100)	
GDPChange VolExports	
GDPChange 1.0000	
VolExports 0.5580 1.0000	

The correlation between export growth and GDP growth across 100 observations is 0.558. This indicates a moderate-strong positive linear relationship between the two variables. Hence, as export volumes increase, GDP increases as well. The correlation analysis supports the regression findings, confirming that export performance is one of the key variables causing GDP fluctuations. While not perfectly linear, the positive correlation suggests that economies with higher export growth also experience stronger GDP performance. This reinforces the hypothesis that export activity plays a key role in driving macroeconomic outcomes.

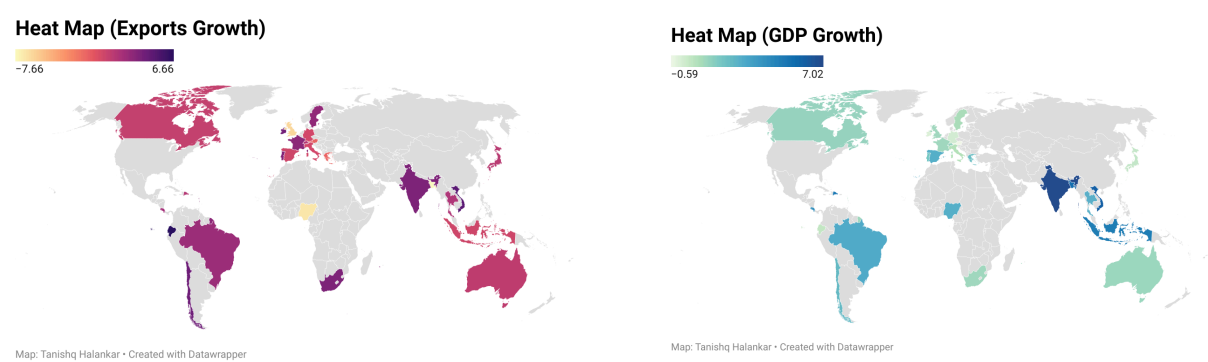


Fig 11 & 12

Conclusion

This paper examined the relationship between export growth and GDP growth across 34 countries over a 20-year period using comprehensive data from the IMF. The analysis confirms the original hypothesis that export growth exerts a statistically significant and positive effect on GDP growth. Although not the only driver of economic performance, exports remain a key pillar of economic growth and development. These findings reinforce traditional trade theory and underscore the value of open-market policies and diversified export strategies in fostering sustained economic growth.

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